

CMPT 354 Mini Project

Step 3: Anomaly and BCNF Analysis

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1 Purpose and Conclusion

This step translates the E/R model from Step 2 into relation schemas and examines the functional dependencies (FDs) that follow from the stated library semantics. A relation is in Boyce–Codd normal form (BCNF) exactly when, for every non-trivial FD $X \rightarrow Y$ that holds in the relation, X is a superkey.

Conclusion. Every relation in the final design is in BCNF. The E/R design already separates facts whose determinants are different, so no further BCNF decomposition is required. Therefore, the final schemas do not permit the insertion, deletion, or update anomalies caused by bad functional dependencies.

2 Relational Translation of the E/R Model

An underlined attribute is the selected primary key. “FK” denotes a foreign key, “AK” denotes an additional candidate key, and “optional” denotes an attribute that may be absent. Foreign-key references are stated after each group so that the schemas remain readable.

2.1 Members and Personnel

MEMBER(memberNo, fullName, email^{AK}, phone, mailingAddress, dateJoined,
status, expiryDate)

EMPLOYEE(employeeNo, fullName, email^{AK}, phone, jobTitle, hireDate,
employmentStatus, supervisorNo^{optional})

In EMPLOYEE, *supervisorNo* is a self-referencing foreign key to *employeeNo*. Its optional value implements the rule that an employee has at most one supervisor.

2.2 Catalogue, Issues, and Lendable Units

CREATOR(creatorID, creatorName)
MATERIALTYPE(typeName, defaultLoanDays)
CATALOGUEITEM(catalogueNo, title, publisher, publicationYear, language,
subject, description, standardID^{optional}, typeName^{FK})
CREATEDBY(catalogueNo, creatorID)
PERIODICALISSUE(catalogueNo, volume, issueNo, publicationDate)
LIBRARYUNIT(unitID, catalogueNo^{FK}, status)
PHYSICALCOPY(unitID, barcode^{AK}, shelfLocation, acquisitionDate, condition)
DIGITALLICENCE(unitID, licenceNo^{AK}, accessURL, licenceLimit)
REPRESENTSISSUE(unitID, catalogueNo^{FK}, volume^{FK}, issueNo^{FK})

CATALOGUEITEM.typeName references MATERIALTYPE. CREATEDBY references CATALOGUEITEM and CREATOR. PERIODICALISSUE references CATALOGUEITEM; its three underlined attributes form one composite key. LIBRARYUNIT references CATALOGUEITEM. The key of each subtype, PHYSICALCOPY and DIGITALLICENCE, is also a foreign key to LIBRARYUNIT. REPRESENTSISSUE references LIBRARYUNIT and the composite key of PERIODICALISSUE. Keeping this optional relationship in a separate table avoids null issue attributes for non-periodical units.

The *standardID* of a catalogue item is optional because some library materials have no ISBN, ISSN, or other standard identifier. When present, it must be unique. Because a classical candidate key must identify every tuple of a relation, an optional *standardID* is not treated as a candidate key or as the determinant of an ordinary FD over all of CATALOGUEITEM. The candidate key is *catalogueNo*. The conditional uniqueness of each non-null *standardID* is a separate integrity constraint for Step 4.

2.3 Loans and Fines

LOAN(loanID, memberNo^{FK}, unitID^{FK}, processedByEmployeeNo^{optional},
borrowedAt, dueAt, returnedAt^{optional})
FINE(fineID, loanID^{FK}, amount, assessedDate, reason, paymentStatus, paymentDate^{optional})

LOAN references MEMBER, LIBRARYUNIT, and optionally EMPLOYEE. FINE references LOAN. The loan foreign key is not unique in FINE, because the E/R model allows one loan to cause zero or more fines.

2.4 Events and Registrations

ROOM(roomNo, roomName, floor, capacity, accessibility, availabilityStatus)
EVENT(eventNo, title, eventType, description, startTime, endTime,
maximumAttendance, status, roomNo^{FK}, organizerEmployeeNo^{FK})
AUDIENCE(audienceName, description)
RECOMMENDEDFOR(eventNo, audienceName)
REGISTRATION(registrationID, eventNo^{FK}, memberNo^{FK}, registrationTime, status, attended)

EVENT references ROOM and EMPLOYEE. RECOMMENDEDFOR references EVENT and AUDIENCE. REGISTRATION references EVENT and MEMBER, and (*eventNo*, *memberNo*) is an additional candidate key. Thus, a member cannot register for the same event twice.

2.5 Proposals, Volunteering, and Help

ITEMPROPOSAL(proposalNo, title, creator, materialType, publicationInformation,
sourceType, dateSubmitted, condition^{optional}, notes, decisionStatus,
decisionDate^{optional}, memberNo^{optional}, reviewerEmployeeNo^{optional})

RESULTSIN(proposalNo, catalogueNo^{AK})

VOLUNTEERAPPLICATION(applicationNo, memberNo^{FK}, applicationDate, areaOfInterest,
availability, skills, status, assignmentNotes^{optional},
reviewerEmployeeNo^{optional}, eventNo^{optional})

HELPREQUEST(requestNo, memberNo^{FK}, category, questionDescription, submittedAt,
priority, status, response^{optional}, completedAt^{optional},
assignedEmployeeNo^{optional})

In ITEMPROPOSAL, *memberNo* and *reviewerEmployeeNo* reference MEMBER and EMPLOYEE. RESULTSIN references ITEMPROPOSAL and CATALOGUEITEM; both attributes are candidate keys because RESULTSIN is one-to-one. VOLUNTEERAPPLICATION references its applicant and optionally its reviewer and supported event. HELPREQUEST references its requesting member and optionally its assigned employee. A proposal's condition may be absent when it is not applicable; volunteer assignment notes may be absent before an assignment is made; and a help-request response and completion time may be absent until the request is resolved.

3 Functional-Dependency Analysis

3.1 Method and Assumptions

For each relation below, a minimal semantic FD basis is given. All other non-trivial FDs follow from this basis by Armstrong's axioms. Writing the minimal basis avoids listing redundant augmented forms such as $\{memberNo, phone\} \rightarrow fullName$ when $memberNo \rightarrow fullName$ already holds.

The following assumptions are part of the data semantics:

- member and employee email addresses are required and unique within their respective relations;
- barcodes and licence numbers are required and unique within their respective subtype relations;
- a standard identifier is unique whenever one is present; because it is optional, this is treated as a conditional uniqueness constraint rather than an ordinary FD over all catalogue-item tuples;
- names, titles, phone numbers, dates, status values, room names, and descriptive fields are not assumed to be unique;
- historical records are retained, so a member, unit, employee, room, or event may occur in many related tuples; and
- no FD is inferred merely from an accidental pattern in the current sample data. An FD is included only if it must hold in every legal database state under the stated business rules.

In the table, $K \rightarrow \text{rest}$ means that K determines every other attribute of that relation. This is a compact representation of the individual non-trivial dependencies having the same determinant.

3.2 Compact FD Basis, Candidate Keys, and BCNF Test

Relation	Minimal non-trivial FD basis	Candidate key(s)	BCNF test
MEMBER	$memberNo \rightarrow rest; email \rightarrow rest$	$\{memberNo\}, \{email\}$	Both determinants are keys.
EMPLOYEE	$employeeNo \rightarrow rest; email \rightarrow rest$	$\{employeeNo\}, \{email\}$	Both determinants are keys.
CREATOR	$creatorID \rightarrow creatorName$	$\{creatorID\}$	The determinant is a key.
MATERIALTYPE	$typeName \rightarrow defaultLoanDays$	$\{typeName\}$	The determinant is a key.
CATALOGUEITEM	$catalogueNo \rightarrow rest$	$\{catalogueNo\}$	The determinant is a key.
CREATEDBY	No non-trivial FDs. Both columns are required to identify one association.	$\{catalogueNo, creatorID\}$	BCNF holds vacuously.
PERIODICALISSUE	$\{catalogueNo, volume, issueNo\} \rightarrow publicationDate$	$\{catalogueNo, volume, issueNo\}$	The determinant is the key.
LIBRARYUNIT	$unitID \rightarrow \{catalogueNo, status\}$	$\{unitID\}$	The determinant is a key.
PHYSICALCOPY	$unitID \rightarrow rest; barcode \rightarrow rest$	$\{unitID\}, \{barcode\}$	Both determinants are keys.
DIGITALLICENCE	$unitID \rightarrow rest; licenceNo \rightarrow rest$	$\{unitID\}, \{licenceNo\}$	Both determinants are keys.
REPRESENTSISSUE	$unitID \rightarrow \{catalogueNo, volume, issueNo\}$	$\{unitID\}$	The determinant is a key.
LOAN	$loanID \rightarrow rest$	$\{loanID\}$	The determinant is a key.
FINE	$fineID \rightarrow rest$	$\{fineID\}$	The determinant is a key.
ROOM	$roomNo \rightarrow rest$	$\{roomNo\}$	The determinant is a key.
EVENT	$eventNo \rightarrow rest$	$\{eventNo\}$	The determinant is a key.
AUDIENCE	$audienceName \rightarrow description$	$\{audienceName\}$	The determinant is a key.
RECOMMENDEDFOR	No non-trivial FDs. Both columns are required to identify one association.	$\{eventNo, audienceName\}$	BCNF holds vacuously.
REGISTRATION	$registrationID \rightarrow rest; \{eventNo, memberNo\} \rightarrow rest$	$\{registrationID\}, \{eventNo, memberNo\}$	Both determinants are keys.
ITEMPROPOSAL	$proposalNo \rightarrow rest$	$\{proposalNo\}$	The determinant is a key.
RESULTSIN	$proposalNo \rightarrow catalogueNo; catalogueNo \rightarrow proposalNo$	$\{proposalNo\}, \{catalogueNo\}$	Both determinants are keys.
VOLUNTEERAPPLICATION	$applicationNo \rightarrow rest$	$\{applicationNo\}$	The determinant is a key.
HELPREQUEST	$requestNo \rightarrow rest$	$\{requestNo\}$	The determinant is a key.

4 Why the Foreign Keys Do Not Create Additional Bad FDs

The arrows in the E/R diagram create several many-to-one relationships. The foreign key on the “many” side is determined by that relation’s key, but the reverse dependency generally does not hold. In particular:

- $memberNo \not\rightarrow loanID$ because one member can have many loans;
- $unitID \not\rightarrow loanID$ because one unit can be borrowed repeatedly over time;
- $loanID \not\rightarrow fineID$ because a loan may cause more than one fine;
- $roomNo \not\rightarrow eventNo$ and $organizerEmployeeNo \not\rightarrow eventNo$ because a room and an employee may be associated with many events;
- $memberNo \not\rightarrow applicationNo$ and $memberNo \not\rightarrow requestNo$ because a member may submit many volunteer applications and help requests; and
- $typeName \rightarrow defaultLoanDays$ is stored only in MATERIALTYPE, not in CATALOGUEITEM. Likewise, $roomNo \rightarrow capacity$ is stored only in ROOM, and member details are stored only in MEMBER.

Consequently, foreign keys do not duplicate descriptive facts in referencing relations. They are inclusion constraints, while the descriptive FDs remain inside the relations whose keys determine those facts.

5 Formal BCNF Proof

Let R be any relation in the final design and let $X \rightarrow Y$ be a non-trivial FD that follows from the library semantics.

1. If R is a regular entity or relationship relation with descriptive attributes, the FD-basis table shows that every minimal determinant is its primary key or an alternate candidate key. Hence X contains a candidate key and is a superkey.
2. If R is CREATEDBY or RECOMMENDEDFOR, there is no non-trivial FD. Each tuple consists only of the two attributes forming the many-to-many association key, so BCNF holds vacuously.
3. If R is RESULTSIN, either attribute determines the other due to the one-to-one constraint. Each determinant is therefore a candidate key.
4. The composite determinant of PERIODICALISSUE is its full key, and the determinant $unitID$ in REPRESENTSISSUE is its key.

Thus, in every possible case, the left side of each non-trivial FD is a superkey. By the definition of BCNF, every relation in the final schema is in BCNF.

6 Anomaly Check

The absence of bad FDs prevents the three anomaly types discussed in class:

- **Update anomalies:** changing a material type’s default loan period, a room’s capacity, a member’s email, or a creator’s name requires one update in its own relation rather than many updates in loan, event, or item tuples.
- **Insertion anomalies:** a new material type, creator, room, audience, member, or future-item proposal can be recorded without first creating a loan, event, registration, or catalogue copy.
- **Deletion anomalies:** deleting a returned loan, cancelled registration, event recommendation, or withdrawn physical copy does not erase the independent member, event, audience, creator, or catalogue facts.

For comparison, a single wide relation containing loan, member, unit, catalogue, and material-type data would contain the bad dependencies

$$\begin{array}{ll} memberNo \rightarrow \text{member details}, & unitID \rightarrow catalogueNo, \\ catalogueNo \rightarrow \{\text{item details}, typeName\}, & typeName \rightarrow defaultLoanDays. \end{array}$$

Those determinants would not be superkeys of the wide relation. The final design avoids this problem by storing the facts in MEMBER, LIBRARYUNIT, CATALOGUEITEM, and MATERIALTYPE, respectively, with LOAN retaining only their foreign keys.

7 Constraints Beyond Functional Dependencies

BCNF prevents anomalies caused by functional dependencies, but it does not by itself enforce every business rule. The following are not bad FDs and will be implemented in Step 4 using keys, foreign keys, CHECK constraints, triggers, transactions, or application logic:

- at most five active loans per member, no borrowing by an inactive member, and borrowing restrictions for overdue items or unpaid fines;
- at most one active loan for a physical copy and at most *licenceLimit* simultaneous loans for a digital licence;
- chronological conditions such as *dueAt* > *borrowedAt* and *endTime* > *startTime*;
- no overlapping events in the same room and attendance not exceeding the event or room capacity;
- total participation requirements such as every catalogue item having at least one creator and every event having at least one audience; and
- conditional uniqueness of each non-null *standardID* and the requirement that the catalogue number in a REPRESENTS ISSUE tuple agree with the catalogue number of its referenced LIBRARYUNIT; and
- conditional rules for accepted proposals, approved volunteer assignments, resolved help requests, and subclass disjointness and completeness.

These integrity constraints complement the BCNF design; they do not require further decomposition.